FIRMWARE STUDY · A POST-BIOS SUITE FOR OMARCHY

Beyond the BIOS

An audit of AM4 BIOS maintenance practices, a map of vendor documentation, and the architecture of a native control suite for Omarchy: the firmware becomes a bootstrap set once, the system becomes the interface.

Ryzen 9 5950X · RTX 3070 · ASUS B450/B550

Omarchy 4.0 · Hyprland · Limine · fwupd

SEPTEMBER 2026 · TECHNICAL REPORT · VOLUME 1

Contents

1. Executive summary

This study began as a field observation: however responsive an Omarchy system may be, it remains constrained by a link inherited from the 1990s, the motherboard BIOS/UEFI. It examines what the firmware of an AM4 platform actually does, what the four major motherboard vendors do — or stop doing — with their BIOS, what their documentation hides, and what the Omarchy project already provides. It concludes with a concrete blueprint: **omarchy-firmware**, a post-BIOS suite that moves 90 percent of BIOS use cases into the system, while scrupulously respecting the conventions of the Omarchy repository.

The central finding has two halves. On one side, the vendor firmware remains irreducible for a minimal foundation: hardware initialisation, DDR4 memory training, the chain of trust. On the other, almost everything that justifies entering the BIOS in daily life — processor settings, fan curves, graphics card behaviour, the boot chain — has a native Linux equivalent, often faster, scriptable and reversible. Between the two lies a tooling gap that no distribution closes end to end today.

455	0	1.2.0.12	281
omarchy-* commands analysed in the repository	AMD CPU tuning tools present in Omarchy	latest AM4 AGESA, deployed by ASUS in August 2026	contract checks now shipped with omarchy-firmware

1.1 The three conclusions of the audit

One — AM4 firmware is not dead; it is frozen in its role. Contrary to a widespread impression, vendors still ship updates in 2026: ASUS delivered BIOS 3644 for the TUF Gaming B550-PLUS on 27 August 2026, carrying AGESA ComboV2 PI 1.2.0.12. But that cadence now covers security and CPU compatibility only. No functional evolution, no interface modernisation, no new tool has enriched the experience in years. The B450 boards, for their part, are largely frozen since 2023 — the MSI B450 Tomahawk MAX has received only beta releases since March 2023.

Two — the security burden has moved into the firmware, not onto the user. Recent flaws — LogoFAIL (2023), Sinkclose CVE-2023-31315 (disclosed 2024), the VU#382314/CVE-2025-14302+ family of December 2025 affecting all four vendors, then the two TPM vulnerabilities of August 2026 (CVE-2026-6726/6727) — impose an update hygiene that the Linux channels do not cover: fwupd and the LVFS catalogue ignore virtually every desktop AM4 motherboard. Updating remains a manual ritual: USB key, EZ Flash, reboots.

Three — Omarchy has already built 80 percent of the foundation; the firmware layer is missing. The repository provides a modern boot chain (UKI + Limine, bootable Snapper snapshots), a fwupd wrapper, idempotent hardware detection and a 455-command CLI router with exemplary conventions. It contains, however, **no** AMD CPU tuning tool, no fan control, no firmware health monitoring — and the Security chapter of its manual never mentions the word firmware. That is precisely the space the blueprint of chapter 8 set out to occupy.

1.2 The three possible trajectories

Three strategies emerge from the study; they are not mutually exclusive. **Trajectory A** — recommended, and detailed in chapter 8 — keeps the current ASUS board and builds the omarchy-firmware suite: the BIOS is visited once a year, for security updates. **Trajectory B** considers a board change to an open-firmware model (ASRock Rack X470D4U, coreboot port, integrated BMC) to host the existing 5950X. **Trajectory C**, further out, anticipates the end of AM4: the MSI PRO B850-P WIFI under Dasharo v0.9.0, the first open-source firmware for a modern AMD desktop platform, released in July 2026. This report provides the facts to decide — and the roadmap to start this week.

1.3 Where the project stands today

This English edition carries one piece of news the French original could only promise: the blueprint of chapter 8 is no longer a blueprint. Between the two editions, the omarchy-firmware suite was specified, built and hardened through five shipped phases, published as an open repository (Cheurteenyt/BIOS-). The current release, 0.6.1, exposes **12 MCP tools** — ten read-only, two reversible — plus a human-only staging layer, a 28-probe behavioural rehearsal backed by a digital twin of the target machine, and the day-0 instruments that turn the first real session into a replay: **capture** (a read-only photograph of the machine) and **rehearse-diff** (the named list of surprises between the twin and reality).

The engineering discipline moved with it. The contract is enforced by 281 automated checks that run on Python 3.11 and 3.13 plus an MCP conformance job; the MCP SDK is pinned to an exact, pre-proven version and a weekly drift canary re-runs the conformance smoke against the floating range a user would actually install; the release payload is a tagged tarball whose SHA256SUMS the installer verifies before anything runs. Nothing here has yet touched the SPI chip — the suite remains strictly read-only or reversible, and the real machine, the AM4 board this study was written for, is scheduled to meet it on 16 September 2026.

2. Context, hardware and methodology

2.1 The target machine

The study is anchored on a precise configuration, representative of the AM4 installed base: a Ryzen 9 5950X (Zen 3, 16 cores / 32 threads), an ASUS motherboard from the B450/B550/X570 family, and an NVIDIA RTX 3070 8 GB (Ampere, GA104 die). This trio imposes three distinct technological constraints: a CPU that depends on AMD's AGESA microcode and on motherboard menus for PBO and the Curve Optimizer; a GPU driven by the proprietary nvidia driver (GSP, nvidia-open-dkms) with its own control levers; and an AMI Aptio firmware whose maintenance depends on the vendor. The system is Omarchy 4.0, DHH's beautiful, fun and agentic Arch Linux, analysed at source level on 9 September 2026.

Component	Model	Relevance to the study
Processor	AMD Ryzen 9 5950X (Zen 3, AM4)	PBO and Curve Optimizer adjustable at runtime via ryzen_smu; AGESA ComboV2 1.2.0.12 required for the 2026 TPM fixes
Motherboard	ASUS B450 / B550 / X570 (AM4 socket, AMI Aptio firmware)	The BIOS remains the only path to RAM timings and boot order; updates flow through USB, not fwupd
Graphics card	NVIDIA RTX 3070 8 GB (GA104, Ampere)	GSP support: nvidia-open-dkms on the Omarchy side; power limit and fan control via nvidia-smi
System	Omarchy 4.0 (Arch, Hyprland, Quickshell)	UKI + Limine chain, Snapper snapshots, omarchy-* CLI router to respect for integration

Table 1 — Target configuration and its constraints.

2.2 The design questions

Before comparing existing solutions, the study asks four design questions — ones the BIOS never had to face for lack of competition. They structure chapters 3 to 7 and justify every choice of the blueprint.

- **What is the irreducible role of a firmware?** Platform wake-up, memory training, chain of trust, recovery. Everything else — menus, optional DXE modules, interfaces — is it avoidable heritage? (chapter 3)
- **Who maintains it, and at what pace?** What the four vendors actually shipped between 2022 and 2026, and what recent flaws impose. (chapter 4)
- **What does the official documentation hide?** What the manuals document, bury or duplicate, and what forums must supply in their place. (chapter 5)
- **Where should each control live: boot-time or runtime?** What Linux can do today — often better, without a reboot — and what must stay in the firmware. (chapters 6 to 8)

2.3 Methodology and sources

The Omarchy analysis rests on a git clone of the official repository (commit 5ead870, 9 September 2026, version 4.0.0.alpha): a reading of the 455 commands in bin/, the AGENTS.md and agents/skills/command-metadata.md conventions, the boot chain (default/limine/limine.conf, omarchy-refresh-limine, omarchy-setup-direct-boot) and the hardware installation scripts. The vendor audit draws on the official support pages (ASUS, MSI, Gigabyte, ASRock), TechPowerUp's AGESA version tracker, AMD security advisories, CERT/CC VU#382314, and research publications (IOActive, Binarly, Eclypsium, Riot Games). The settings map cross-references the official BIOS manuals with documented vendor forum threads (ROG Forum, MSI forum) where the manuals fall short.

Two methodological limits must be stated. First, the BIOS date audit relies on support pages as of collection time: those pages evolve, and a vendor may ship an unlisted beta. Second, the exact behaviour of a setting (Memory Context Restore in particular) depends on the board revision and the AGESA version; the findings of chapter 5 are documented trends, not per-model guarantees. Neither limit affects the blueprint or the roadmap, both designed to be robust to the heterogeneity of the installed base.

3. Anatomy of AM4 firmware: what a BIOS really does

3.1 The four irreducible functions

Reducing a firmware to its interface is the most common mistake. In reality, four functions cannot live anywhere else. **Platform reset** first: between the power button and the first pixel of the loader, the firmware runs the SEC and PEI phases of UEFI, wakes the CPU to AMD microcode's rhythm and applies AGESA — the proprietary block AMD ships to vendors to initialise DDR4 memory, PCIe links and power planes. **Memory training** next: on every boot without a saved context, the memory controller re-derives its timings through an iterative process that costs 30 to 90 seconds on AM4, depending on DIMMs per channel and their speed. **The chain of trust** last: Secure Boot, platform keys, fTPM — the hardware anchor the OS assumes healthy when it wakes.

The fourth function is the least visible: **recovery**. A healthy firmware must be able to re-flash a corrupted chip without a working CPU (BIOS FlashBack), boot from a backup (dual BIOS) or reset the CMOS. It is this last-resort contract that justifies the blueprint's caution: everything the omarchy-firmware suite does can be undone from the firmware itself. Everything else — the hundreds of DXE modules, the mouse-driven interfaces, the licence screens, the pre-OS network assistants — is heritage, not contract.

3.2 The boot chain, compared

The diagram below compares three boot chains. The first is the default vendor path: it traverses the full set of UEFI phases, the costliest of which remains memory training. The second is what Omarchy installs today: the vendor firmware is kept, but the visible part shrinks to a three-second Limine menu, Tokyo Night themed, pointing at a Unified Kernel Image — a single binary bundling kernel, initramfs and a signed command line. The third is the blueprint's target: an EFI entry created by efibootmgr that skips the menu, for a hands-off boot.

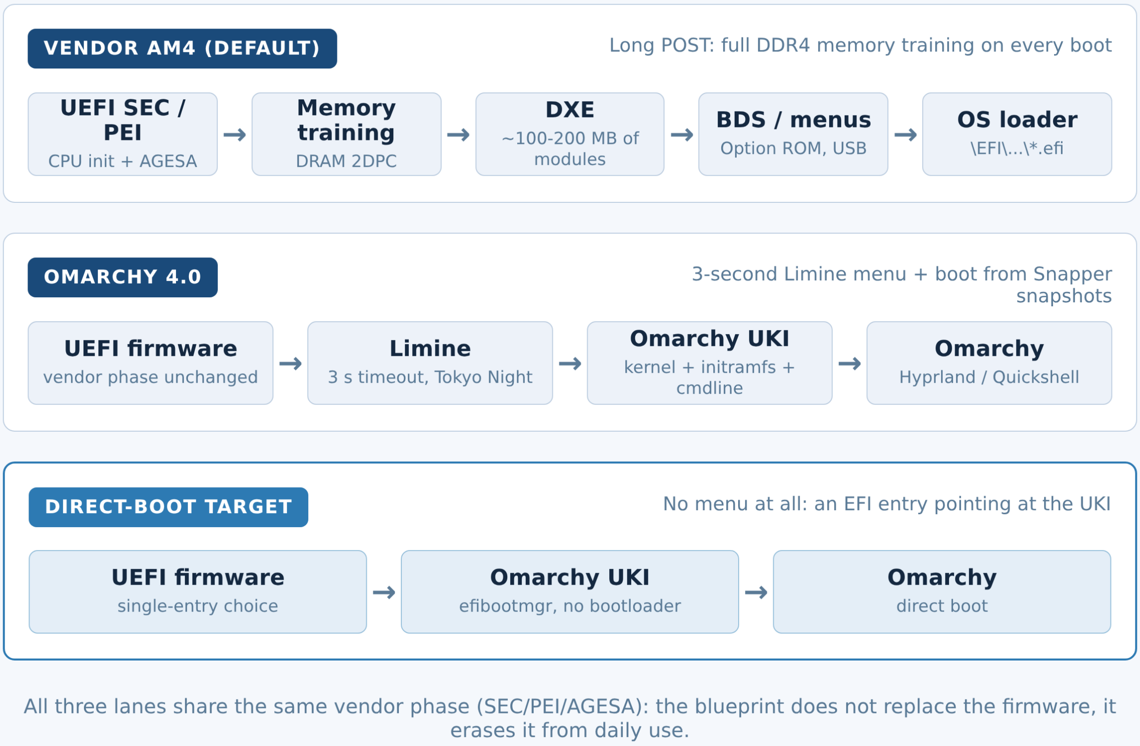

Figure 1 — Boot chains compared: vendor default, Omarchy 4.0 today, the blueprint's direct-boot target.

The essential point of the diagram is that the three lanes share the same vendor phase: the blueprint does not replace the firmware, it erases it from daily use. That is exactly the lesson of the open-firmware community revisited in chapter 7: hardware init is an implementation detail; boot policy and settings belong to the system.

3.3 Why AM4 POST drags: MCR and Power Down Enable

AM4's slow boot has a precise cause and a documented — but poorly surfaced — remedy. Without optimisation, the DDR4 controller reruns full memory training on every boot. The **Memory Context Restore** (MCR) setting asks the firmware to save the training context and reuse it; **Power Down Enable** (PDE) completes the mechanism on the DIMM power side. On most B550 boards, enabling MCR + PDE cuts the POST from several dozen seconds to under ten. The setting carries a known trade-off: after a memory configuration change or an AGESA update, a restored context can be invalid — vendor forums then recommend a forced retrain or a CMOS reset. This kind of critical, badly documented setting is the perfect illustration of the work chapter 5 maps and that the blueprint's boot module automates downstream (verification, not configuration).

> **Takeaway.** AM4 firmware keeps a monopoly on four functions: platform wake-up, memory training, the chain of trust, recovery. Everything that happens after those four — that is, the daily experience — is territory Linux can claim. The blueprint of chapter 8 is the act of claiming it.

4. Audit of vendor BIOS maintenance practices

4.1 The AGESA rhythm: security without product

Since 2022, AGESA ComboV2 (the second generation, for Zen 2 and later) has followed a revealing trajectory. Version 1.2.0.7 (mid-2022) fixes the infamous fTPM stuttering bug that haunted Ryzen 3000/5000 under Linux — a textbook firmware-OS interaction, completed by a kernel fix in 2023. The 1.2.0.8 line (2023) carries AMD's response to Sinkclose, the SMM vulnerability disclosed by IOActive in August 2024 and eighteen years old. Versions 1.2.0.C then 1.2.0.F (2025) continue the hygiene, and 1.2.0.12 — shipped to vendors around May 2026 — fixes the two TPM flaws referenced as CVE-2026-6726 and CVE-2026-6727, which reach from Ryzen 3000 to Ryzen AI.

That rhythm tells one story: AMD maintains AM4 as a **security asset**, no longer as a product. Each AGESA release carries fixes or new-CPU compatibility; none carries an experience improvement. On the vendor side, deployment is a long tail: Gigabyte shipped 2E45 with 1.2.0.12 on the B550 Aorus Elite AX V2 in summer 2026; ASUS shipped 3644 on the TUF Gaming B550-PLUS on 27 August 2026; ASRock had followed 1.2.0.F in early September 2025 on the B550 Steel Legend; MSI, on B450, stopped at March 2023 betas. Table 2 summarises the landscape.

Board (AM4 socket)	Last BIOS observed	Date	Embedded AGESA	Support status
ASUS TUF Gaming B550-PLUS	3644	27/08/2026	ComboV2 PI 1.2.0.12	Active — security
ASUS TUF Gaming B550-PLUS (history)	3636 / 3634 beta	27/01/2026 / 24/09/2025	1.2.0.12 / 1.2.0.B	Semi-annual cadence
Gigabyte B550 Aorus Elite AX V2	2E45	summer 2026	1.2.0.12 + TPM fix CVE-2026-6726/6727	Active — security
ASRock B550 Steel Legend	10.x (Instant Flash)	08/09/2025	1.2.0.F + updated fTPM	Regular support
MSI B450 Tomahawk MAX	7C02v3G1 (beta)	09/03/2023	ComboAm4v2 PI 1.2.0.8	Frozen — no stable release since 2023
MSI B450 Tomahawk (non-Max)	2024-2025 updates	2024-2025	1.2.0.F + TPM firmware	Late fixes, long tail

Table 2 — Last BIOS observed per board (vendor support pages, September 2026).

4.2 The vulnerabilities that impose hygiene

The 2023-2026 period turned AM4 firmware into an attack surface in its own right. LogoFAIL (December 2023) showed that a single malicious image file on the EFI partition could execute code in the DXE phase; vendor fixes only began arriving en masse in April 2024, bundled with the AGESA of the moment. Sinkclose (CVE-2023-31315, IOActive, August 2024) let a ring-0 attacker install themselves at ring -2, beneath the OS — invisible to antivirus; its AGESA fix took months to propagate board by board. In December 2025, CERT/CC published VU#382314: UEFI modules from all four vendors prevent proper IOMMU initialisation, opening DMA attacks at the very beginning of boot (CVE-2025-14302 on Gigabyte, CVE-2025-14304 on ASRock), discovered by Riot Games researchers. Finally, in August 2026, AMD confirmed two flaws in the TPM reference code (CVE-2026-6726, CVSS 8.5; CVE-2026-6727, CVSS 8.3) allowing forged attestation outputs — fixed only by AGESA 1.2.0.12.

Flaw	Disclosed	Main impact	Fix and deployment
fTPM stuttering (AMD advisory)	2022	Periodic micro-freezes under Linux (TPM RNG)	AGESA 1.2.0.7 (2022) + kernel fix (2023)
LogoFAIL (Binarly)	Dec. 2023	Pre-boot code execution via malicious DXE images	Vendor fixes from April 2024, long per-board tail
Sinkclose — CVE-2023-31315 (IOActive)	Aug. 2024	Ring 0 to SMM escalation (ring -2), stealthy persistence	AGESA 1.2.0.8+ line; rollout spread over 2024-2025
VU#382314 — CVE-2025-14302/14304 (Riot Games, CERT/CC)	Dec. 2025	Uninitialised IOMMU: DMA attacks in early boot	Dedicated BIOS per board at ASUS, Gigabyte, MSI, ASRock
CVE-2026-6726 / 6727 (AMD TPM)	Aug. 2026	Forged TPM attestation (CVSS 8.5 / 8.3), Ryzen 3000 to Ryzen AI	AGESA 1.2.0.12 (OEM May 2026); ASUS and Gigabyte Aug. 2026

Table 3 — Timeline of firmware vulnerabilities affecting AM4, 2022-2026.

4.3 fwupd and LVFS: the Linux channel that misses motherboards

Omarchy already ships the right tool: **omarchy update firmware** installs fwupd, copies the EFI capsule update executable, then delegates to fwupdmgr refresh and fwupdmgr update. The problem sits upstream: the LVFS catalogue contains only what manufacturers choose to deposit. Laptops, docks, SSDs and peripherals are well served — ASUS was the first motherboard vendor to publish there in 2020 — but desktop AM4 motherboards are massively absent, and public fwupd project discussions confirm that each manufacturer decides model by model, for support and liability reasons. Practical consequence for the target machine: Omarchy's command will show a sober no-devices-found or list only peripheral devices, and every BIOS update will follow the manual ritual: download the .CAP from the ASUS page, drop it on a FAT32 key, reboot into EZ Flash, wait, reconfigure. That ritual is exactly what the blueprint's audit module proposes to frame — version detection, correlation with the latest known AGESA, a post-flash verification checklist — without ever pretending to replace the vendor flow.

> **Reading the audit.** AM4 vendors maintain the contractual security minimum — slowly, board by board, never enriching the tool. Waiting for a better BIOS from them is a strategic dead end: the innovation has moved. It now lives in the OS, and that is where Omarchy can seize it.

5. Mapping the BIOS documentation

5.1 The paradox of the pre-OS interface

The BIOS is the only major component whose interface never had to face comparison. It runs before any OS, competes with nothing, and its user sees it a few minutes per year — in moments of tension, where a mistake can leave the machine unbootable. That absence of competitive pressure produces documentation to match: the official manuals describe every option in one neutral sentence (Enables or Disables...) without ever explaining interactions, recommended Linux values, or the critical settings. The settings that matter most for an Omarchy desktop — Memory Context Restore, Power Down Enable, the fTPM's position, the double PBO menu — are precisely the ones the manuals skim or ignore, leaving vendor forums to serve as the real documentation.

5.2 The matrix of critical settings

Table 4 crosses the settings that matter for the target machine with how the four vendors expose them — and with whether Linux can drive them. Three observations stand out. First, the marketing settings (Resizable BAR, Above 4G) are documented and accessible everywhere: they sell boards. Next, the experience settings (MCR, PDE) exist everywhere but are buried at varying depths, absent from manuals, and sometimes tied to other options in undocumented ways — on ASUS, the ROG forum points to an MRC Fast Boot option you have to search for. Finally, the double PBO menu: every BIOS exposes both a vendor menu and a hidden AMD Overclocking menu, with different values and behaviours — the MSI forum explicitly advises against touching the second. One setting, two interfaces, zero documentation of the difference.

Setting	Exposure in AM4 BIOSes	Documented?	Drivable from Linux?
Memory Context Restore + Power Down Enable	Buried (DRAM / Tweaker / OC); on ASUS tied to MRC Fast Boot; depth varies by vendor	No — forums only	No (boot-time) — but verifiable downstream
PBO + Curve Optimizer	Double menu: vendor menu (OC) and hidden AMD Overclocking menu — diverging values	Partly — third-party guides	Yes — pbo2-tuner / ryzen_smu, live on Zen 3
Above 4G Decoding / Re-Size BAR	Standard menu, recently favourable defaults	Yes	No (boot-time) — state readable via dmesg
fTPM (dTPM)	Standard menu everywhere; performance implications undocumented	Partly	No (boot-time) — version readable via sysfs
Fan curves (Q-Fan / Smart Fan)	Vendor graphical interface, CPU/board sensors only	Yes	Yes — hwmon / lm-sensors, including GPU-linked
GPU: power limit, fans, clocks	Absent from the BIOS by design	n/a	Yes — nvidia-smi / coolbits (RTX 3070)
Boot order / Secure Boot	Standard menu	Yes	Partly — efibootmgr from the OS; keys via sbctl

Table 4 — Critical AM4 settings: vendor exposure, documentation, Linux equivalent.

5.3 What this implies for the project

This mapping has a direct consequence for the blueprint's architecture. The boot-time settings (MCR, PDE, Above 4G, fTPM) cannot be driven from Linux: the suite will not set them — it will **document and verify** them: an audit module that reads the real state via sysfs, dmi and dmesg, compares it against the embedded knowledge base's recommendations, and produces a pre-flash checklist. Conversely, everything runtime — PBO, Curve Optimizer, fans, GPU — moves wholly to the OS side: that is the suite's functional core. The asymmetry is deliberate: the firmware stays a verified bootstrap, Linux becomes the single interface. Here we find, applied to a workstation, the mechanism/policy separation that has structured open firmware for fifteen years.

6. Inside the Omarchy project

6.1 What the repository reveals

Reading the repository (commit 5ead870, 9 September 2026, version 4.0.0.alpha) reveals a project of rare discipline for a community tool. The repo produces two packages: **omarchy** — binaries, install scripts, migrations, themes and the Quickshell desktop — and **omarchy-settings**, everything that must be in place before the user is created, including the full boot and snapshot history (mkinitcpio, limine-entry-tool drop-ins, snapper template). Three layers populate the home: seed via /etc/skel, finalise via omarchy-provision-user, and an explicit resync to overwrite configs. The main CLI router, bin/omarchy, scans the metadata of the 455 commands in bin/ — every script declares its summary, group, arguments and sudo need through normalised comments. That convention is the blueprint's natural entry point: any compliant command is automatically listed, documented and routed.

```
# omarchy:summary=Apply a CPU profile (eco / stock / perf)
# omarchy:args=[eco|stock|perf|status]
# omarchy:examples=omarchy firmware cpu profile perf
# omarchy:requires-sudo=true
```

The extract above shows the target convention: four comment lines suffice for a new command to appear in Omarchy's help, menus and tests. Prefixes are normalised (hw- for detection that returns an exit code, refresh-, setup-, toggle-, update-), the $OMARCHY_PATH environment is guaranteed by the session, and the privilege-escalation policy is documented in the repo's skills.

6.2 The boot chain and the existing flows

Omarchy 4.0's boot chain is already that of a post-BIOS system: a UKI (omarchy_linux.efi in /boot/EFI/Linux/), a Limine configured by default (3-second timeout, Tokyo Night palette, interface_branding Omarchy Bootloader), synchronised with Snapper snapshots so an earlier system state can be booted. Two commands frame the mechanism: **omarchy-refresh-limine** resets limine.conf from the repo defaults and reruns limine-update and limine-snapper-sync; **omarchy-setup-direct-boot** creates (or removes) an EFI entry pointing straight at the UKI. That second script contains a detail worth underlining for the target machine: it **explicitly refuses** to create a direct entry on American Megatrends firmwares, judging that they do not handle custom EFI entries sanely. Since virtually every ASUS AM4 board runs an AMI Aptio firmware, full direct boot remains a target to validate carefully on this machine, entry by entry — the blueprint's boot module includes precisely that validation (entry creation, three reboot cycles, a rollback test).

The update flow shows the same rigour: omarchy-update checks free disk space, takes a Snapper snapshot before operations, keeps the machine awake, logs everything to /tmp/omarchy-update.log and offers a firmware step delegated to fwupd (see chapter 4 on LVFS limits). Hardware detection is idempotent (omarchy-apply-hardware, replayable) and NVIDIA boards are identified by reading sysfs — vendor

0x10de, display class — with automatic switching to nvidia-open-dkms when GSP support is available, which is the case for the RTX 3070's Ampere, with early KMS configured in mkinitcpio.

6.3 The voids: where the blueprint settles

Void identified in the repository	Proof of absence	Blueprint answer (chapter 8)
No AMD CPU tuning (PBO, Curve Optimizer, profiles)	grep zenstates / pbo2 / ryzen_smu / curve optimizer: 0 hits in bin/, config/, install/, docs/, manual/	cpu module: declarative profiles, pbo2-tuner + amd_pstate
No fan control	No fancontrol / hwmon / pwm reference outside a backlight fix	fans module: hwmon curves, GPU link, systemd daemon
No firmware health monitoring	Manual chapter 48-security.md: zero occurrences of firmware / BIOS / Secure Boot / TPM	audit module: AGESA version, Secure Boot state, CVE checklist
Boot: no cautious direct-boot tooling for AMI	omarchy-setup-direct-boot refuses AMI with no alternative	boot module: cycle-based validation, readable limine.conf
NVIDIA GPU: installed but no runtime control	install/hardware/nvidia.sh installs packages, no power/fan settings	gpu module: power limit, fan curve, per-use profiles

Table 5 — Omarchy's five functional voids and the blueprint's answer.

These voids are not criticisms: they are reasonable scope choices for a young project that targets laptops and mainstream desktops first. But for the target machine — an AM4 workstation with a discrete GPU — they are the five daily friction points. The omarchy-firmware suite is designed to close exactly that gap, integrating with the existing mechanisms (Quickshell menus via omarchy.menu routes, the update step, Snapper snapshots as the safety net) rather than inventing a parallel architecture.

7. Lessons from the open-firmware community

7.1 Mapping the ecosystem

The community never built a better BIOS: it built a layered architecture, and fifteen years of projects sketch it. **coreboot** replaces proprietary initialisation with open code — keeping the AGESA blob on AMD as an acknowledged compromise. **LinuxBoot** replaces UEFI's DXE phase with a Linux kernel that kexecs into the final system: the firmware becomes Linux, with shell, network and native debugging — the Open Compute server approach. **Heads** and PureBoot add verification: measured and TPM-verified boot, GPG-protected secrets, tamper detection. **Dasharo** industrialises it all as a distribution: its v0.9.0 of July 2026 for the MSI PRO B850-P WIFI is the first open-source firmware for a modern AMD desktop platform. **System76** open-sources its machines' firmware via coreboot + EDK2, **OpenBMC** provides the server world's out-of-band management, **oreboot** rewrites coreboot in Rust, and **fwupd/LVFS** standardises OS-side updates.

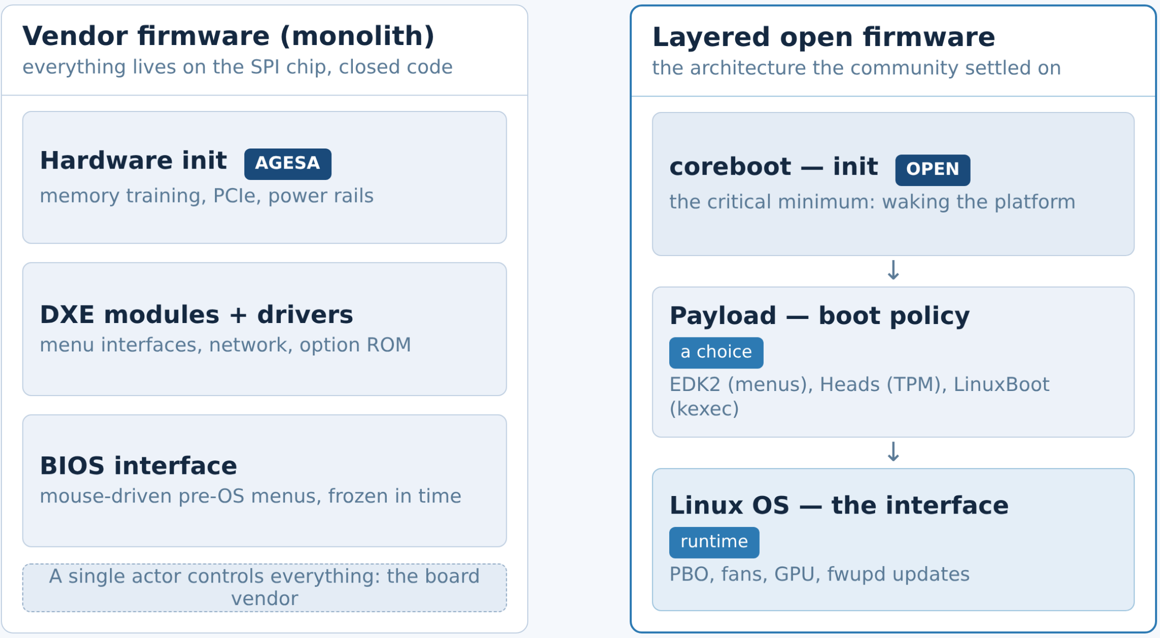

Figure 2 — The vendor-firmware monolith versus the layered architecture of open firmware.

Project	Guiding idea	Relevance to AM4 desktop
coreboot	Open hardware init, payload of choice (EDK2, Heads, LinuxBoot)	Ports limited to server boards (ASRock Rack X470D4U/X570D4U); no consumer B450/B550 port
LinuxBoot	A Linux kernel as the firmware phase, kexec into the OS	Requires a coreboot port; inspiring for design, out of reach on the current board
Heads / PureBoot	Measured and verified boot (TPM, GPG), anti-tampering	Specific target hardware (ThinkPad, NitroPad); principles transferable (post-flash verification)
Dasharo	Industrialised coreboot distribution, Secure Boot, tracked releases	AM5 only (MSI PRO B850-P WIFI, v0.9.0 July 2026) — the trajectory-C replacement
System76 Open Firmware	coreboot + EDK2 open source, vertically integrated	Organisational model: firmware as a maintained product with regular releases
OpenBMC	Linux management processor (remote console, sensors)	Available via server boards (IPMI); useful in trajectory B
fwupd / LVFS	Firmware updates from the OS, signed capsules	Already integrated in Omarchy; desktop AM4 coverage near zero (ch. 4)

Table 6 — The open-firmware ecosystem and its applicability to the target machine.

7.2 The design principle to keep

All these approaches share one intuition: **separate mechanism from policy**. The mechanism — waking the platform, loading a boot image — is an implementation detail nobody wants to see. The policy — what to boot, how to cool, when to update, how to verify — belongs to the system, to its tools, to its versionable configuration files. Vendor firmware fuses both into a closed monolith; open firmware separates them into interchangeable layers. For the target machine, where replacing the firmware is out of reach (chapter 4), the lesson applies differently: keep the vendor mechanism, but reclaim the whole policy. That is exactly the blueprint's brief: an omarchy firmware router compliant with the repo's conventions, declarative versioned modules, snapshot-reversible changes, and documentation that lives in the OS rather than in a pre-OS PDF.

> **The decisive shift.** The BIOS is not an interface: it is a loader. The interface of a modern computer is its system. The whole study fits inside that reversal — and the next chapter is its translation into code.

8. Blueprint: the omarchy-firmware suite

8.1 The brief in one sentence

Build, for Omarchy, the layer the vendor BIOS will never provide: an **omarchy firmware** router declined into five modules — audit, boot, cpu, fans, gpu — where every repeatable setting becomes a declarative command, versionable, snapshot-reversible, and automatically exposed in Omarchy's help and menus through the repository's normalised metadata. The BIOS is visited only for what it alone can do: memory training, the chain of trust, recovery, and the annual security flash.

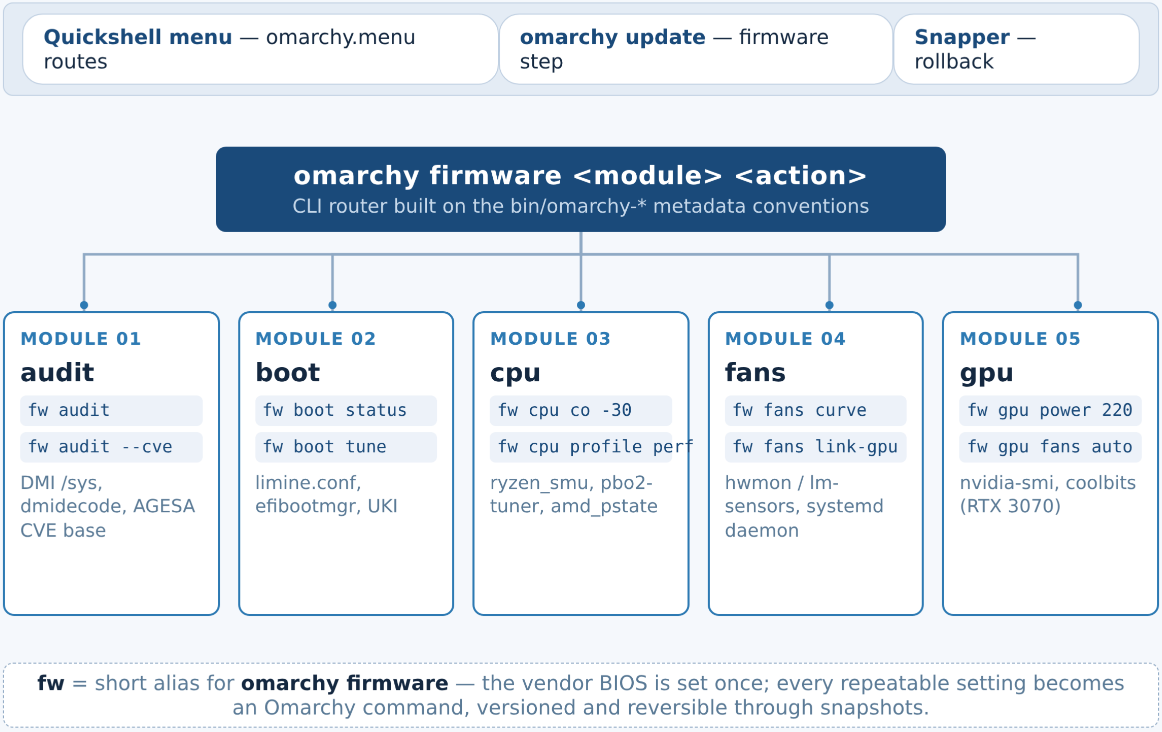

Figure 3 — Module map of the omarchy-firmware suite: CLI router, five modules, integration rails.

8.2 Module specification

Module	Target commands	Data sources	Design notes
audit	omarchy firmware audit / audit --cve	DMI (/sys/class/dmi), dmidecode, dmesg, embedded AGESA CVE base	Read-only; compares installed AGESA against latest known, Secure Boot/fTPM state; outputs a pre-flash checklist
boot	omarchy firmware boot status / boot tune	limine.conf, efibootmgr, UKI /boot/EFI/Linux, snapper list	Chain status, timeout, entries; tune adjusts timeout and default entry; cycle-based direct-boot validation (AMI)
cpu	omarchy firmware cpu co -30 / cpu profile perf	ryzen_smu, pbo2-tuner (Zen 3), amd_pstate (EPP)	Declarative eco/stock/perf profiles; live per-core Curve Optimizer; stability check (stress + temperature watch)
fans	omarchy firmware fans curve / fans link-gpu	hwmon, lm-sensors, dedicated systemd daemon	Curves per sensor, including GPU temperature; silent/perf presets; minimum-step guards
gpu	omarchy firmware gpu power 220 / gpu fans auto	nvidia-smi, coolbits (X config), nvidia-open-dkms	Power limit, locked clocks, fan curve; per-use profiles (gaming, compute, silence); RTX 3070 as target

Table 7 — Functional specification of the five modules.

Three principles run through the five modules. **Declarative first**: settings live in profile files (TOML or YAML under $OMARCHY_PATH), never in a machine's implicit state — a profile is a file, a file is versionable, a version is reversible. **Reversibility next**: every state-changing command first takes a lightweight Snapper snapshot (as omarchy-update does) and can restore the default via a --reset argument. **Compliance last**: # omarchy:summary/args/examples/requires-sudo metadata on every script, normalised prefixes, $OMARCHY_PATH guaranteed, the repo's sudo/pkexec policy respected — which makes the suite an immediate citizen of the ecosystem: contextual help, Quickshell menus, acceptance tests.

8.3 Omarchy integration, module by module

Integration does not stop at the CLI router. The Quickshell **menu** exposes an omarchy.menu firmware route with the common actions (profiles, status, curves) — the repo's IPC pattern (omarchy menu summon firmware) allows it without shell-side specific code. The **omarchy update** command gains a post-pacman step: refresh the fwupd cache, flag any AGESA drift the audit module detects. The **manual** receives a chapter 52, Firmware — like the existing 51, written for the end user — documenting MCR/PDE, fTPM, PBO and the flash ritual, closing the gap mapped in chapter 5. Finally, the **embedded CVE base** is a simple JSON versioned in the repository (flaw families, fixing AGESA versions), updated with every AMD advisory — the same mechanics as a package, with no external service.

8.4 The roadmap in three phases

Phase	Content	Acceptance criteria
Phase 1 — foundation (weeks 1-2)	omarchy firmware router + audit module + boot module (status/tune); manual Firmware chapter; CVE JSON base	omarchy firmware audit produces a correct report on the target machine; boot status reflects limine.conf and efibootmgr; commands listed in omarchy help
Phase 2 — runtime CPU/GPU (weeks 3-5)	cpu and gpu modules: eco/stock/perf profiles, live Curve Optimizer, NVIDIA power limit and fans	Profile switch without reboot; 30-minute stress stable under perf; --reset rollback effective; RTX 3070 capped under 220 W
Phase 3 — cooling and consolidation (weeks 6-8)	fans module (hwmon curves, GPU link); Quickshell menu + omarchy update step integration; full documentation	GPU-active curve thermally validated; menus functional; usage feedback integrated; repository published

Table 8 — Roadmap, milestones and acceptance criteria.

The schedule is deliberately conservative: each phase ships a state usable on its own, and the priority order follows daily pain — knowing where your firmware stands (audit), mastering your boot (boot), then reclaiming the settings the BIOS holds hostage (cpu, gpu, fans). Phase 1 alone already transforms the experience: the user knows, in one command, whether their machine is up to date, healthy and compliant with the expected configuration.

8.5 Status: the blueprint has shipped

This English edition can replace the future tense with the past. Between the French original and this translation, all five phases of the extended roadmap shipped as the omarchy-firmware repository (Cheurteenyt/BIOS-), release 0.6.1. The scope grew beyond the original five modules: the audit became a twelve-tool MCP surface (ten read-only T0 tools, two reversible T1 tools, plus a human-only T2 staging layer refused to agents), the risk matrix of chapter 9 became a structural test discipline (281 checks, two Python versions, MCP conformance), and two day-0 instruments were added — a digital-twin rehearsal machine and the capture/rehearse-diff pair that turns the first real session into a scored replay.

Phase	Shipped as	Delivered evidence
P1 — T0 foundation	0.1.0 — 4 read-only tools + skill + access journal	10 state questions answered without any write; refusals proven by test
P2 — thermal engine	0.2.x — signature engine, active probe, baseline, doctor timer	8 scenarios named one by one on fixtures; measured frugality
P3 — full T0 + T1 HAL	0.3.x — storage/GPU/RAM/settings diagnostics + cpu.epp.set + fans.curve.set	177 checks, MCP 11-tool conformance, verified rollback, mechanical curve guards
P4 — supervised loop	0.4.x — cve.watch, human-gated T2 staging, report digest, weekly timer	246 checks, 12 scenarios, MCP 12-tool conformance; the 5-day criterion awaits the machine
P5 — twin + day-0 kit	0.5.x/0.6.x — TWIN-1 profile, 28-probe rehearsal, rehearse-diff, capture; pinned CI + drift canary; SHA256-verified install	281 checks green on py3.11/3.13; rehearsal green on the twin; release v0.6.0 roundtrip-verified

Table 8b — The shipped state of the roadmap as of 10 September 2026.

9. Risks, limits and the validation plan

9.1 The honest limits of the project

A study that does not say what its subject cannot do is not a technical report. Six limits structure omarchy-firmware's real perimeter. RAM timings and boot order remain unreachable from the OS: the suite documents them, does not set them. pbo2-tuner targets the Zen 3 family — exactly the target machine's 5950X, but not Zen 1/2, which will go through amd_pstate profiles alone. Since the RTX 3070 runs nvidia-open, fine fan curves go through coolbits (X configuration) rather than a pure-system setting: a documented, reversible compromise. The BIOS flash remains a vendor ritual: the suite frames it (checklist, verifications) and does not automate it — deliberately, given the brick risk. Memory Context Restore stays a boot-time setting with its known stability trade-off: the suite verifies its coherence after a flash, never toggles it remotely. Finally, full direct boot on AMI firmware contradicts omarchy-setup-direct-boot's caution: it will be validated cycle by cycle before being offered, never enabled by default.

Risk	Likelihood	Impact	Mitigation
Instability after an aggressive Curve Optimizer	High (by nature)	Crashes under load, wasted time	-5 steps, 30-min stress validation, documented --reset, prior snapshot
AGESA regression after flash (invalid MCR)	Low	Long POST or boot failure	Audit module's post-flash checklist; CMOS-clear procedure in the manual
Badly calibrated hwmon fan curve	Medium	Localised overheating	Enforced minimum steps, hysteresis, integrated btop monitoring
Direct EFI entry swallowed by AMI firmware	Medium	Machine boots to the UEFI shell	3-cycle validation, Limine entry kept as fallback (Omarchy's current scheme)
Dependency on third-party tools (pbo2-tuner, ryzen_smu)	Medium	Breakage on a kernel change	Documented version pinning, graceful detection, amd_pstate degradation
Divergence from Omarchy 4.x's fast evolution	Medium	Changed CLI conventions or menus	Upstream tracking, acceptance tests, migrations in the repo's style

Table 9 — Risk matrix and matching umbrellas.

Two mitigations were added after the French edition, both now structural. The supply chain: the MCP SDK is pinned to one exact version proven against the conformance smoke before the pin was written, and a weekly scheduled canary re-runs that smoke against the floating range a user would install — if the ecosystem drifts, a red job notices before any machine does. The distribution: releases are tagged, the payload is a tarball published with its SHA256SUMS, and the installer verifies the checksum before staging anything, echoing the tag and digest so the session log can quote them. Neither mitigation existed in the blueprint; both were paid for by incidents the project chose to treat as teachers.

9.2 The validation plan

Every shipped feature follows the same four-step protocol, in the spirit of Omarchy's graphical acceptance tests. **Measured initial state**: temperatures, frequencies, POST time, power draw captured before any change and archived. **Single change**: one command, one setting, one Snapper snapshot before applying — never two modifications at once. **Instrumented validation**: timed stress under supervision (btop, hwmon sensors, systemd journal) with explicit failure thresholds. **Compared rollback**: final measurements against the initial state, kept as history — the profile file is the regression test. The 0.5.x phase added the decisive extension: the whole behavioural contract (28 probes, from tier refusals to MCP negotiations) now runs unattended against a digital twin of the target machine, so the real machine only ever confirms what the twin already proved.

10. Conclusion and roadmap

The founding intuition survives the audit, sharpened and tooled. An AM4 board's BIOS is neither dead nor harmless: it has become a security module maintained at the contractual minimum, whose interface and documentation interest no one — not even their authors. Opposite it, Omarchy had already built most of a post-BIOS system: UKI + Limine chain, bootable snapshots, a disciplined CLI router, a fwupd wrapper. Between the two lay a precise tooling gap — five voids, no more, no less — that the omarchy-firmware suite set out to close without ever touching the SPI chip or contradicting the vendor firmware in its irreducible functions.

That gap is now closed in code, not only on paper: five phases shipped, 281 contract checks green across two Python versions, a twelve-tool MCP surface conformant and pinned, a digital twin that rehearses the whole contract without hardware, and a day-0 kit that will turn 16 September 2026 into a replay rather than a discovery. The remaining phase is the one no repository can ship: the machine itself. Its first session — capture, rehearsal, diff, and the named list of real-world surprises — is the study's final examination, and its result will feed the next iteration of the tool, not the next rewrite of the plan.

The long trajectory is already written by the 2026 landscape. When the 5950X gives way to a recent platform, open firmware will pose its question in new terms: the MSI PRO B850-P WIFI under Dasharo — the first open-source desktop AMD firmware, published in July 2026 — proves that a fitting BIOS, maintained by a community, is no longer an enthusiast project but a product. By then the suite will have played its double role: making the current machine immediately good, and preparing the conceptual vocabulary — profiles, audit, verification, reversibility — that will transfer intact to tomorrow's open firmware. The BIOS we deserved was never to be found at the vendors: it was to be built, one command at a time, in the OS we already chose.

Sources

Repository and project documentation

Omarchy (Basecamp/DHH) — official repository, commit 5ead870, 9 September 2026: AGENTS.md, agents/skills/command-metadata.md, docs/file-layout.md, bin/omarchy-update-firmware, bin/omarchy-setup-direct-boot, bin/omarchy-refresh-limine, default/limine/limine.conf, install/hardware/nvidia.sh, manual/30-updates.md, manual/48-security.md. https://github.com/basecamp/omarchy

Limine Boot Protocol and configuration: https://github.com/limine-bootloader/limine (CONFIG.md), as embedded in default/limine/limine.conf.

The tool this study produced

omarchy-firmware — repository built from this study's blueprint, releases 0.1.0 to 0.6.1 (September 2026): lib/firmware_hal (12-tool MCP surface, digital twin, 28-probe rehearsal, rehearse-diff, capture), tests/test_suite.py (281 checks), .github/workflows/ci.yml (pinned SDK, drift canary), install.sh (--from, SHA256SUMS verification), docs/first-run.md (the day-0 protocol). https://github.com/Cheurteenyt/BIOS-omarchy-firmware — release v0.6.0 and its assets: omarchy-firmware-0.6.0.tar.gz + SHA256SUMS (e11484f2...), the canonical day-0 payload. https://github.com/Cheurteenyt/BIOS-/releases/tag/v0.6.0

AM4 firmware, AGESA and security

AMD — Security Advisory, SMM Lock Bypass (Sinkclose, CVE-2023-31315), August 2024; IOActive, DEF CON 32 presentation; CSO Online (9 August 2024) and Dark Reading (12 August 2024) coverage.

AMD — Trusted Platform Module (TPM) Reference Code Errata, CVE-2026-6726 (CVSS 8.5) and CVE-2026-6727 (CVSS 8.3), August 2026; Tom's Hardware, videocardz.com and igorslab.de coverage (12-14 August 2026).

CERT/CC — Vulnerability Note VU#382314, Vulnerability in UEFI firmware modules prevents IOMMU initialization, 17 December 2025; CVE-2025-14302 (Gigabyte, advisory TVN-202512003), CVE-2025-14304 (ASRock); discoveries by Riot Games; BleepingComputer coverage (19 December 2025).

Binarly — LogoFAIL: advisory December 2023; Gigabyte Security Advisory (15 December 2023); Tom's Hardware, AMD motherboard partners start rolling out BIOS updates with LogoFAIL bugfix, 10 April 2024.

TechPowerUp — AMD AGESA Versions, ComboV2 revision tracker (including 1.2.0.12), accessed September 2026.

Vendor support pages, accessed September 2026: ASUS TUF GAMING B550-PLUS (BIOS 3644, 27/08/2026, AGESA ComboV2 PI 1.2.0.12; 3636, 27/01/2026); MSI B450 TOMAHAWK MAX (7C02v3G1 beta, 09/03/2023, AGESA 1.2.0.8); Gigabyte B550 AORUS ELITE AX V2 (2E45, AGESA 1.2.0.12 + TPM fixes); ASRock B550 Steel Legend (Instant Flash 08/09/2025, AGESA 1.2.0.F).

AMD — fTPM stuttering advisory (2022) and AGESA ComboV2 PI 1.2.0.7 fix; Phoronix, Linux Fix Coming For Recent Stuttering On AMD Ryzen Due To fTPM RNG, 14 February 2023; Linux kernel Bugzilla #217461 (2023).

Eclypsium — AMD Processors Vulnerable to Malicious Microcode, 6 February 2025.

Phoronix — ASUS Offers First Motherboard Firmware Update Via LVFS + Fwupd, 9 November 2020; fwupd GitHub discussion #8125 (21 November 2024) on motherboard coverage.

Open firmware and hardware trajectories

coreboot — official documentation (doc.coreboot.org) and supported distributions (Heads, EDK2); coreboot.org/pipermail on the retirement of AMD 14h/15h/16h platforms (7 November 2022).

LinuxBoot — linuxboot.org (FAQ), LWN.net, LinuxBoot: Linux as firmware, 7 March 2018; github.com/linuxboot/linuxboot.

Dasharo — 3mdeb, MSI PRO B850-P WIFI Dasharo v0.9.0 Released, 30 July 2026; docs.dasharo.com; Tom's Hardware, First open-source firmware for AM5 officially launches, 2 August 2026.

System76 — github.com/system76/firmware-open; opensource.com, System76 introduces laptops with open source BIOS, 5 November 2019.

OpenBMC — openbmc.org and github.com/openbmc; ASRock Rack X470D4U/X570D4U (product pages, Ryzen 5000 support, IPMI).

ASUS ROG Forum — Memory Context restore and Power Down thread (August 2025); MSI forum — MSI Memory Context Restore AM5 Settings and Power Down Enable (27 January 2025); Tom's Hardware, How to use Precision Boost Overdrive and Curve Optimizer, 11 March 2026.